

BIOLOGY

All Important Scientist

Important Year

Full Form



Tarun Sir

Class 11th (Scientist and Their Discoveries NCERT New Edition)

1. **Carolus Linnaeus:** Binomial Nomenclature
systema naturae- title of his publication
Two Kingdom Classification
Artificial system
2. **Aristotle:** Scientific system, plants-herb, shrub and tree and animal with RBC and without RBC
3. **R.H. Whittaker (1969):** Five Kingdom classification
4. **Bentham and Hooker** Natural classification system
5. **Dmitri Ironowsky (1892):** Bacterial proof filter
6. **M. W. Beijerinck (1898):** Discover virus, Contagium vivum fluidum
7. **K. M. Stanley (1935):** crystallised and crystals contain largely of proteins
8. **T. O. Diener (1971):** Discovered 'viroids'
9. **1960: Katherine Esau:** Published "Anatomy of Seed Plants."
10. **1954: Ramachandran:** Triple helical model of collagen, published
11. **Anton von Leeuwenhoek** First saw and described a live cell.
12. **Robert Brown:** Discovered nucleus
13. **Matthias Schleiden (1838)** All plants are composed of different kinds of cells which form tissues
14. **Theodore Schwann:** Studied -animal cells → thin outer layer → today known as the plasma membrane. (1839)
15. **Schwann:** Proposed the hypothesis that the bodies of animals and plants are composed of cells and product of cells.
16. **Schleiden and Schwann:** Formulated the cell theory.
17. **Rudolf Virchow (1855):** Modified cell theory, Omnis cellula e cellulae
18. **Singer and Nicolson:** Fluid mosaic model of plasma membrane (1972)
19. **Camillo Golgi (1898):** Discovered golgi apparatus
20. **George Palade (1953):** Observed ribosomes as dense particles under electron microscope.
21. **Flemming** Gave the name 'chromatin' to the material of nucleus.
22. **Joseph Priestley (1774):** Bell jar experiment
23. **Jen Ingenhousz:** Showed sunlight is essential to plant process.
24. **Julius von Sachs (1853)** Give evidence production of glucose when plants grow in special structure
25. **T.W. Engelmann:** Described the first action spectrum of chl a which is similar to action spectrum of chl a and chl b
26. **Van Niel** inferred that the O₂ evolved by green plant comes from H₂O, not from CO₂.
27. **Calvin :** Use C¹⁴ to discovered first product of CO₂ Fixation in Algae and also work out complete pathway
28. C₄-pathway was discovered by **Hatch & Slack**
29. **Blackman (1905):** Law of limiting factors
30. **Embeden, Mayerhoff, Parnas (EMP)** Gave the scheme of glycolysis
31. **Hans Kurb:** Citric acid cycle or TCA cycle
32. **F. W. Went :** Isolated auxin from tip of coleoptiles of oat seedling.
33. **E. Kurosawa (1926):** Discovery of 'gibberellic acid'
34. **Charles Darwin and:** Observed that coleoptiles of Canary grass responded to unilateral Francis Darwin illumination by growing towards the light source.
35. **F. Skoog and co worker** Intermodal segments of tobacco stems the callus grow when auxins the nutrients medium was supported with → extract of vascular tissue, yeast extract, coconut milk or DNA.
36. **Skoog & Miller** identified, crystallised & termed Kinetin.
37. **Cousins** – Ripened organs promote ripening of banana
38. Good humor hypothesis was disproved by **William Harvey**
39. Blood circulation was discovered by- **William Harvey**

Class 12th (Scientist and Their Discoveries NCERT New Edition)

1. **1951:** Family planning programme started in India
2. **1971:** Govt. of India legalized MTP.
3. **Gregor Mendel (1856-1863)**- Seven years - Laws of Inheritance
4. **R.C punett** – British – punett square
5. **Hugo de Vries, Correns, Tschermak -1900:** Rediscovered Mendel's work
6. **Walter Sutton and Theodore Boveri (1902):** Proposed chromosomal theory of Inheritance
7. Genetic map discovered by **Alfred Sturtevent**
8. **T.H morgan:** Experimental verification of the chromosomal theory of inheritance and linkage
9. **Henking -1891:** Trace a specific nuclear structure all through spermatogenesis in a few insects
10. **Langdon Down (1866):** Described Down's syndrome
11. **Fredich Miescher(1869):** Identified DNA as an acidic substance in the nucleus of a cell.
12. **James Watson and Francis Crick (1953):** Proposed double helical structure and semi- conservative nature of DNA.
13. **Franklin and Wilkins :** X-ray crystallography data
14. **Erwin Chargaff:** Ratios b/w adenine and thymine, and cytosine and guanine is constant and equals to one.
1926- Quest to determine genetic material
15. **Frederick Griffith (1928):** Transforming Principle
16. **O. Avery, C. Macleod and M. McCarthy (1933- 44):** Biochemical characterization of Transforming Principle
17. **M. Meselson and F. Stahl (1958):** Experimental Proof for semi-conservative nature of DNA replication.
18. Unequivocal proof that DNA is the genetic material came from the experiments of **Hershey & Chase (1952).**
19. Replication scheme was given by **Watson & Crick**
20. Central dogma of biology was given by **Crick**
21. **Taylor- 1958:** Semiconservative DNA replication – in vicia faba- using radioactive thymidine
22. **George Gamow:** Three nucleotide code for one amino acid
23. **Har Gobind Khurana:** Developed chemical method to synthesise RNA molecules with defined combination of bases (homopolymers and copolymers)
24. **Marshall Nirenberg:** Developed cell- free system for protein synthesis- Deciphering of genetic code
25. **Severo ochoa:** Discovered enzyme polynucleotide phosphorylase
26. **Francois Jacob and Monod :** Gave the concept of Lac operon Jacques Monod
27. Automated DNA sequencers worked on the principle of a method developed by **Frederick Sanger**
28. **Frederick Sanger** developed the method for determination of amino acids sequences in proteins.
29. **Alec Jeffreys:** DNA fingerprinting
30. **Early Greek Thinkers:** 'Theory of panspermia'
31. **Charles Darwin:** Based on observations made during voyage ship -H. M.
S. Beagle-concludes - life forms share similarities not only among themselves - also with life forms - existed millions of years ago.
32. **Louis Pasteur:** Life existing from pre-existing life
1953: Miller experiment - Methane, ammonia, hydrogen & water vapour.
33. **Oparin and Haldane:** Proposed 'Chemical evolution of life'
34. **Charles Darwin:** Natural selection and 'survival of the fittest'
35. **Alfred Wallace:** Co-developed 'theory of natural selection' with Darwin
36. **Ernst Haeckel:** Embryological support for evolution
37. **Lamarck:** Proposed that 'evolution of life forms is driven by use and disuse of organs'.
38. **Karl Ernst Von Baer :** Disapproved embryological support for evolution
39. **Thomas Malthus:** His works on populations influenced Darwin
40. **Hugo de Vries:** Worked on 'Evening Primrose' to put forth the idea of mutations
41. **Hardy- Weinberg:** Proposed that frequency of occurrence of alleles of a gene or a locus remain fixed and same through generations.
42. **1938:** Coelocanth fish caught in South Africa
43. **1891:** Fossil discovered in Java – Homo erectus.
44. **1981:** AIDS was first reported.
45. **Stanley Cohen and Herbert Boyer (1972) :** Constructed the first recombinant DNA (rDNA)
46. **1963:** Two enzyme responsible for restricting growth of bacteriophage in E. coli were isolated.

47. **Eli Lilly (1983)**: Synthesised human insulin
48. **1997**: 1st transgenic cow- Rosie. It produces human protein enriched milk (2.4 grams per litre). The milk contain the human alpha- lactalbumin.
49. **1990**: 1st clinical gene therapy.
50. **1997**: An American company got patent rights on Basmati Rice.
51. **1937**: **Ramdeo Misra** obtained Ph.D. in Ecology from Leeds university.
52. **Edward Wilson** – Give term biodiversity , sociobiologist
53. **Robert May**: Placed the global species diversity at about 7- million.
54. **Alexander Von Humboldt** : Species area relationship
55. **David Tilman**: Long term External plot experiment
56. **Paul ehrlich** Rivet popper hypothesis
57. "Competitive Exclusion principle" was given by **Gause**
58. Resource partitioning was given by **Mac Arthur**
59. **1992**: The Earth summit: Rio de Janeiro conservation of biodiversity
60. **2002** -world summit – Johannesburg ,south Africa , Sustainable Development

All Full forms of Biology – class 11th and 12th

1. **ICBN**: International Code for Botanical Nomenclature
2. **ICZN**: International Code of Zoological Nomenclature
3. **CDRI**:-Central Drug Research Institute (Lko.)
4. **PSTD**:-Potato Spindle tuber Disease
5. **PPLO**:-Pleuro Pneumonia Like Organisms
6. **GFR**:-Glomerular Filtration Rate
7. **JGA**:-Juxta-Glomerular Apparatus
8. **ANF**:-Atrial-Natriuretic factor
9. **GnRH**:-Gonadotrophin Releasing Hormone
10. **GH**:-Growth Hormone
11. **PRL**:-Prolactin
12. **TSH**:-Thyroid Stimulating Hormone
13. **ACTH**:-Adreno Corticotrophic Hormone
14. **MSH**:-Melanocyte Stimulating Hormone
15. **ADH**:-Vasopressin/Anti-diuretic Hormone
16. **T3**:-Triiodothyronine
17. **T4**:-Tetraiodothyronine or thyroxine
18. **TCT**:-Thyrocalcitonin
19. **PTH**:-Parathyroid hormone
20. **CCK**:-Cholecystokinin
21. **GIP**:-Gastric Inhibitory Peptide
22. **LHC**:-Light Harvesting Complex
23. **PGA**:-Phosphoglyceric acid
24. **OAA**:-Oxalo acetic acid
25. **RuBisCO**:-Ribulose bisphosphate carboxylase-oxygenase.
26. **NAD**:-Nicotinamide Adenine Dinucleotide (Coenzyme).
27. **COA**:-Coenzyme A
28. **TCA**:-Tricarboxylic Acid Cycle or Citric Acid Cycle.
29. **ETC**:-Electron Transport Chain
30. **RO**:-Respiratory Quotient
31. **PGRs**:-Plant Growth Regulators
32. **NAA**:-Napthalene Acetic Acid
33. **2,4-D**:-2,4-Dichlorophenoxy acetic acid
34. **GA3**:-Gibberellic acid.
35. **PMC**:-Pollen mother cell/microscope mother cell.
36. **MMC**:-Megaspore mother cell.
37. **PEN**:-Primary Endosperm Nucleus.

38. **hCG**:-Human Chorionic Gonadotrophin
39. **hPL**:-Human Placental Lactogen
40. **RCH**:-Reproductive & Child Health Care.
41. **STDs**:-Sexually Transmitted Diseases
42. **VD**:-Venereal disease.
43. **RTI**:- Reproductive Tract Infections
44. **WHO**:- World Health Organisation
45. **MMR**:- Maternal Mortality Rate
46. **IMR**:- Infant Mortality Rate
47. **IUDs**:- Intra Uterine Devices
48. **OCPS**:- Oral Contraceptive Pills
49. **MTP**:- Medical Termination of Pregnancy
50. **ART**:-Assisted Reproductive Technologies
51. **ZIFT**:-Zygote Intra-fallopian transfer
52. **GIFT**:-Gamete Intra-fallopian transfer
53. **IUT**:-Intra Uterine Transfer
54. **IUI**:-Intra Uterine Insemination
55. **AIT/AT**:-Artificial Insemination Technique
56. **ICSI**:-Intra cytoplasmic sperm injection
57. **IVF**:-In-vitro fertilisation
58. **ET**:-Embryo transfer
59. **SCA**:-Sickle cell Anaemia
60. **PKU**:-Phenylketonuria
61. **NHC**:-Non-histone Chromosomal proteins
62. **snRNA**:-Small Nuclear RNA
63. **hnRNA**:-Heterogenous Nuclear RNA
64. **SRNA**:-Soluble RNA
65. **UTR**:-Untranslated Regions
66. **HGP**:-Human Genome Project
67. **ELSI**:-Ethical, Legal & Social Issues
68. **BAC**:-Bacterial Artificial Chromosome
69. **YAC**:-Yeast Artificial Chromosome
70. **ESTS**:-Expressed Sequence Tags
71. **SNPs**:-Single Nucleotide Polymorphism
72. **VNTR**:-Variable Number of Tandem Repeats
73. **IARI**:-Indian Agricultural Research Institute (New Delhi)
74. **CMI**:-Cell-Mediated Immunity
75. **MALT**:-Mucosal Associated Lymphoid Tissue
76. **AIDS**:-Acquired Immuno Deficiency Syndrome
77. **HIV**:-Human Immuno Deficiency Virus
78. **ELISA**:-Enzyme Linked Immuno Sorbent Assay
79. **NACO**:-National AIDS Control Organisation
80. **NGOs**:- Non-Governmental Organisation
81. **C-onc**:-Cellular Oncogenes
82. **CT**:-Computed Tomography
83. **MRI**:-Magnetic Resonance Imaging
84. **LSD**:-Lysergic acid diethyl amide
85. **SCP**:-Single Cell Protein
86. **LAB**:-Lactic Acid Bacteria
87. **STPS**:-Sewage Treatment Plants
88. **BOD**:-Biochemical oxygen Demand
89. **DO**:-Dissolved Oxygen
90. **KVIC**:-Khadi & Village Industries Commission
91. **Bt**:-Bacillus thuringiensis.
92. **NPV**:-Nucleopolyhedro virus

- 93. **IPM**:-Integrated Pest Management
- 94. **EcoRI**:-Escherichia coli RY13.
- 95. **Ori**:-Origin of Replication
- 96. **pBR322**:-Plasmid
- 97. **Ti Plasmid**:-Tumor Inducing Plasmid
- 98. **Tag**:-Thermus aquaticus
- 99. **PCR**:-Polymerase Chain Reaction
- 100. **GMO**:-Genetically modified Organisms
- 101. **RNAi** :-RNA interference
- 102. **ADA**:-Adenosine deaminase
- 103. **GEAC**:-Genetic Engineering Approval Committee.
- 104. **NPP**:- Net Primary Productivity
- 105. **GPP**:-Gross Primary Productivity
- 106. **GFC**:-Grazing food chain
- 107. **DFC**:-Detritus food chain
- 108. **IUCN**:-International Union for Conservation of Nature & Natural Resources